Weekly Assignment: Strong Induction, Recursive Sequences, & Recurrences (Epp 5.4, 5.6, 5.7, & 5.8)

Complete each of the following problems. Show all work and justify your answers.

Problem 1 (Epp 5.4 #3). Suppose that $c_0, c_1, c_2, \ldots$ is a sequence defined as follows:

$$c_0 = 2, \quad c_1 = 2, \quad c_2 = 6, \qquad c_k = 3\, c_{k-3} \text{ for every integer } k \geq 3.$$

Prove that c_n is even for each integer $n \geq 0$.

Problem 2 (Epp 5.4 #15). Define the "sum" of one integer to be that integer itself. Use strong mathematical induction to prove that for every integer $n \geq 1$, any sum of n even integers is even.

Problem 3 (Epp 5.6 #4). Find the first four terms of the recursively defined sequence:

$$d_k = k\,(d_{k-1})^2, \text{ for every integer } k \geq 1, \qquad d_0 = 3.$$

Problem 4 (Epp 5.6 #12). Let $s_0, s_1, s_2, \ldots$ be defined by the formula $s_n = \dfrac{(-1)^n}{n!}$ for every integer $n \geq 0$. Show that this sequence satisfies the following recurrence relation for every integer $k \geq 1$:

$$s_k = \frac{-\, s_{k-1}}{k}.$$

Problem 5 (Epp 5.7 #2 b). The formula

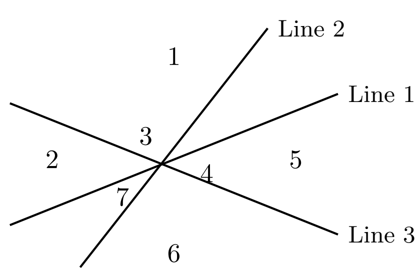
$$1 + r + r^2 + \cdots + r^n = \frac{r^{n+1} - 1}{r - 1}$$

is true for every real number $r \neq 1$ and every integer $n \geq 0$. Use this fact to find a closed-form formula for:

$$3^{n-1} + 3^{n-2} + \cdots + 3^2 + 3 + 1,$$

where n is an integer with $n \geq 1$.

Problem 6 (Epp 5.7 #52). A single line divides a plane into two regions. Two lines (by crossing) can divide a plane into four regions; three lines can divide it into seven regions (see figure below). Let P_n be the maximum number of regions into which n lines divide a plane, where n is a positive integer.



a. Derive a recurrence relation for P_k in terms of P_{k-1}, for each integer $k \geq 2$.

b. Use iteration to guess an explicit formula for P_n.

Problem 7 (Epp 5.8 #2). Which of the following are second-order linear homogeneous recurrence relations with constant coefficients? For each, briefly explain why or why not.

a. $a_k = (k-1)\,a_{k-1} + 2k\,a_{k-2}$

b. $b_k = 2\,b_{k-1} + 7\,b_{k-2}$

c. $c_k = 3\,c_{k-1} + 1$

d. $d_k = 3\,d_{k-1}^2 + d_{k-2}$

e. $r_k = r_{k-2} - 6\,r_{k-3}$

f. $s_k = s_{k-1} + 10\,s_{k-2}$